Australian Artificial Intelligence Institute

Artificial Intelligence for Scientometrics

Yi Zhang
Associate Professor
Australian Artificial Intelligence Institute | School of Computer Science
Faculty of Engineering & Information Technology
University of Technology Sydney (UTS)

Dr Yi Zhang

Associate Professor

Australian Artificial Intelligence Institute (SIG Coordinator of Responsible AI) & **School of Computer Science** (Responsible Academic Officer)

Faculty of Engineering & Information Technology
University of Technology Sydney (UTS)

PhD in Management Science & Engineering (BIT, 2016) & Software Engineering (UTS, 2017)

ARC DECRA Fellow (2019)

The Australian's Research Award (2023)

- Australia's Research Field Leader in Library & Information Science

Research Interest (LLM + Graph Learning + Bibliometrics)

- *Technology & innovation management*
  - Technological forecasting and assessment
- *Science of science, and team science*
  - Responsible teaming recommendation
- *Information behaviors and business information systems*
  - Human-AI collaboration

Professional Services

- **Executive Editor**
  - *Technological Forecasting & Social Change*
- **Specialty Chief Editor**
  - *Frontiers in Research Metrics and Analytics*
- **Associate Editor**
  - *Scientometrics*
  - *IEEE Transactions on Engineering Management*
  - *International Journal of Computational Intelligent Systems*

What is AI?

- A field in computer science
- Develop methods **enabling machines to perceive their environment** and **uses learning and intelligence to take actions** that maximise their chances of achieving defined goals

- **AI vs. "X"**
  - Computer Science, Information Technologies,
  - Information Systems, Digital Technologies
  - Automation, etc.
- **Perceive**, **Learn**, and **Maximise**
- Automation or Information Systems
  - Repeat pre-defined tasks without perceiving data and learning activities.
- Traditional Business Intelligence or Data Analytics
  - Perceive data and "identify" patterns and relationships, without learning/predicting activities
- ChatGPT
  - Perceive large-scale historical data
  - Learn relationships and patterns
  - Maximise the accuracy in answering questions

How AI helps Scientific Research

- Human-centric AI
  - **Perceive Data** (Summarisation)
    - Know what we have already known
  - **Learn Patterns** (Prediction)
    - classification, recommendation, etc.
    - Entity recognition and relationship discovery

- Human and AI collaboration
  - **Inference:** Know what we do not know
  - “Effectively” maximise the chance
    - ChatGPT’s result manipulation (**hallucination**)

Responsible AI for ST&I

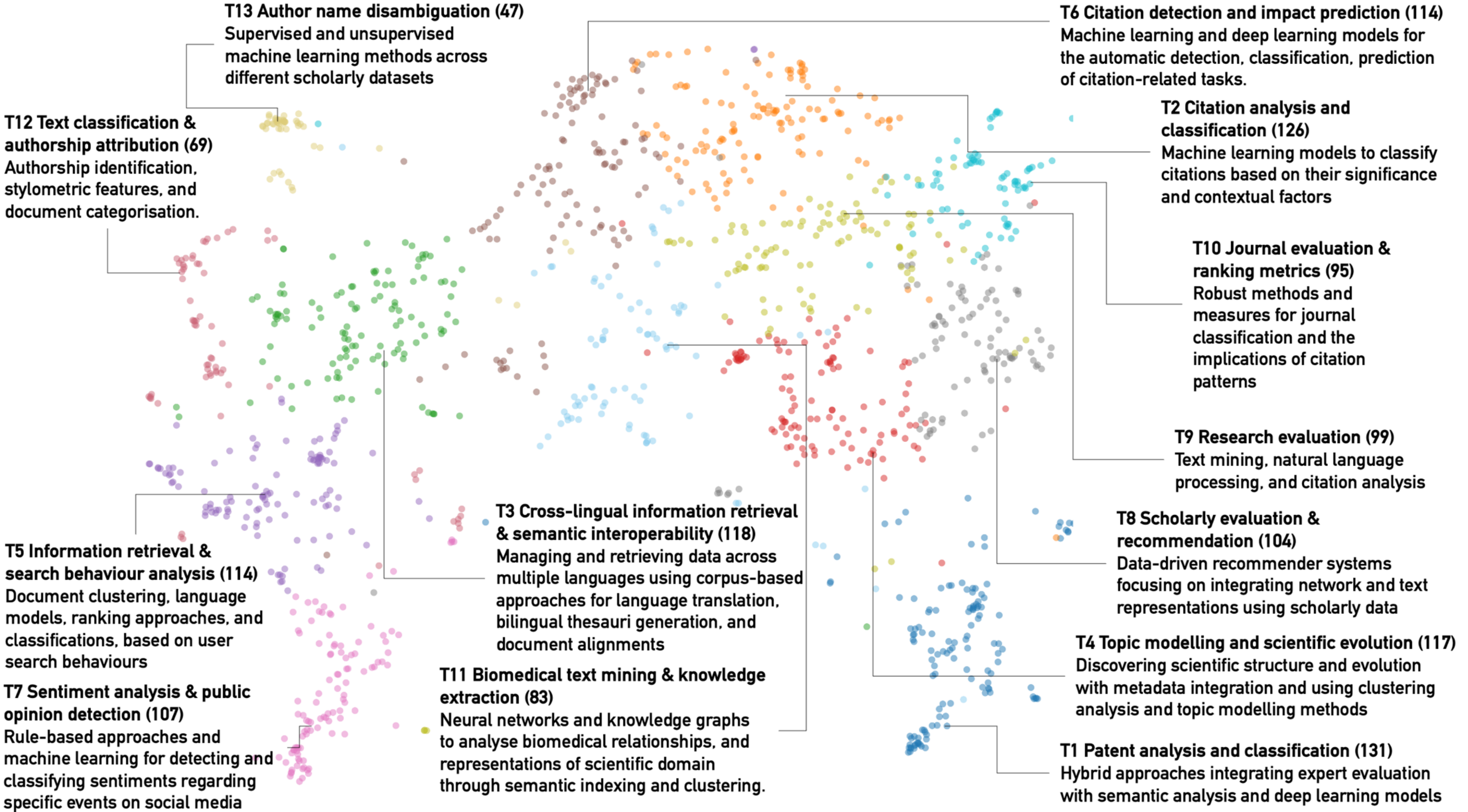
- **Reliability & Reproducibility** [Fundamental Principles]
  - Robust theoretical foundation and validating methodologies through rigorous experiments.
- **Explainability & Transparency** [Advanced Concepts]
  - Interpretable methodological design and compelling method selection with details.
  - Evidence-based result delivery
- **Inclusiveness** [Finishing Touches]
  - Diversity, equity, inclusion, and belonging (DEIB) for underrepresented individuals, population groups, and communities.

Zhang, Y. (2023). Responsible models and indicators: challenges from artificial intelligence. *Frontiers in Research Metrics and Analytics*, *8*, 1305692.

Zhang, Y. et al. 2025. Artificial intelligence for scientometrics: A survey. *Scientometrics*, *to appear*.

AI for SCIM: What have we done

- **1324 Articles**
  - JASIST, SCIM, JOI, QSS, and IP&M
  - Till 31/07/2024
- **Classification**
  - GPT 4o-mini
  - AI4SCIM or SCIM with AI
- **Topic Model**
  - BERTopic and labelled with GPT 4o-mini
- **Four Cohorts**
  - Research evaluation
  - Scientometric IR
  - Patentometrics4STI
  - Behaviour analysis on social media



AI for SCIM: Challenges

- How to represent and analyse complicated real-world ST&I scenarios

Current AI Solutions

- Machine learning & deep learning
- Knowledge representation and reasoning

HOWEVER

Data scalability vs. insufficiency

- The right way to analyse the right data

A socio-technical system: Measures and relations

- Interactions among ST&I entities, science and public policy, public sentiments, etc.

An evolving complex ST&I system

- Dynamics of entities and their relationships

What is intelligent bibliometrics

Bibliometrics/Scientometrics/Informetrics

The development and application of **computational models** incorporating **artificial intelligence and/or data science technologies** with **bibliometric indicators** for handling real-world ST&I issues

Intelligent Bibliometric Networks

- **Heterogeneous bibliometric networks**
  - Bi-layer/tri-layer networks, e.g., co-term network + co-authorship network
  - Knowledge graphs, e.g., Microsoft Academic Graph, and biomedical graphs
- **Graph mining and graph representation learning on bibliometric networks**
  - Meta-path-based analysis, e.g., link prediction
  - Graph neural networks, e.g., graph transformer

Tracing Topical Relationships

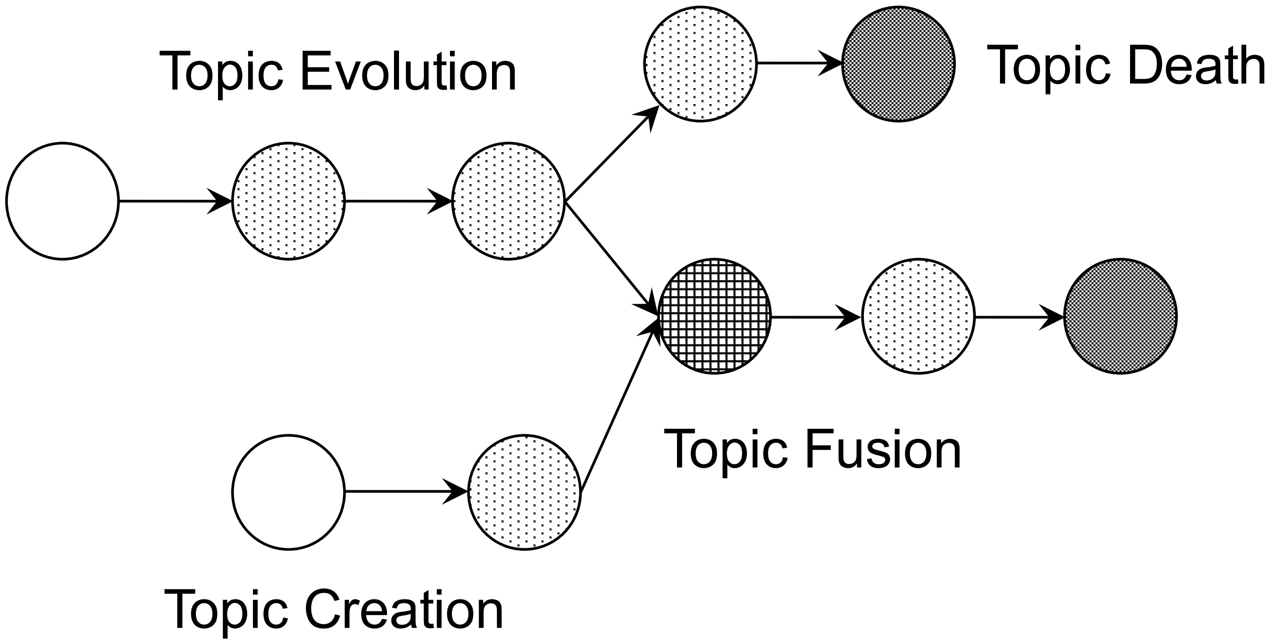
Zhang, Y., Zhang, G., Zhu, D., & Lu, J. (2017). Science evolutionary pathways: Identifying and visualizing relationships for scientific topics. *Journal of the Association for Information Science and Technology*, 68(8), 1925-1939.

Tracing Topical Relationships
- Streaming data analytics-based relationship discovery

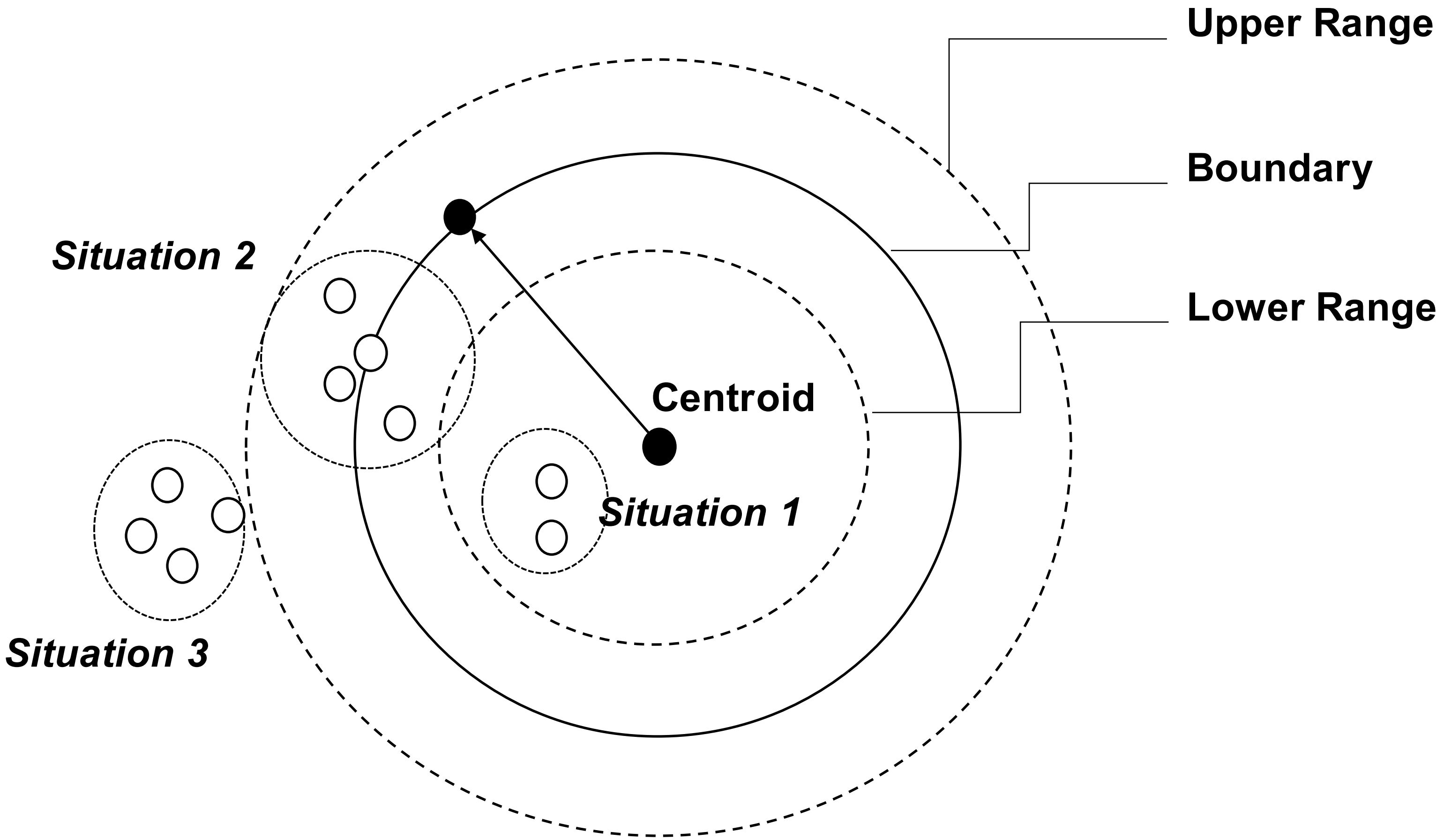
Basic idea:

Topic is changing over the time, e.g., the main content of the topic “data mining” is changing from “database management” and “data warehouse” decades ago to “streaming data analytics” and “machine learning” these days.

How can we grasp such potential change?



Tracing Topical Relationships
- Streaming data analytics-based relationship discovery



- Situation 1 – normal topic
- Situation 2 – topic change
- Situation 3 – novelty or noise

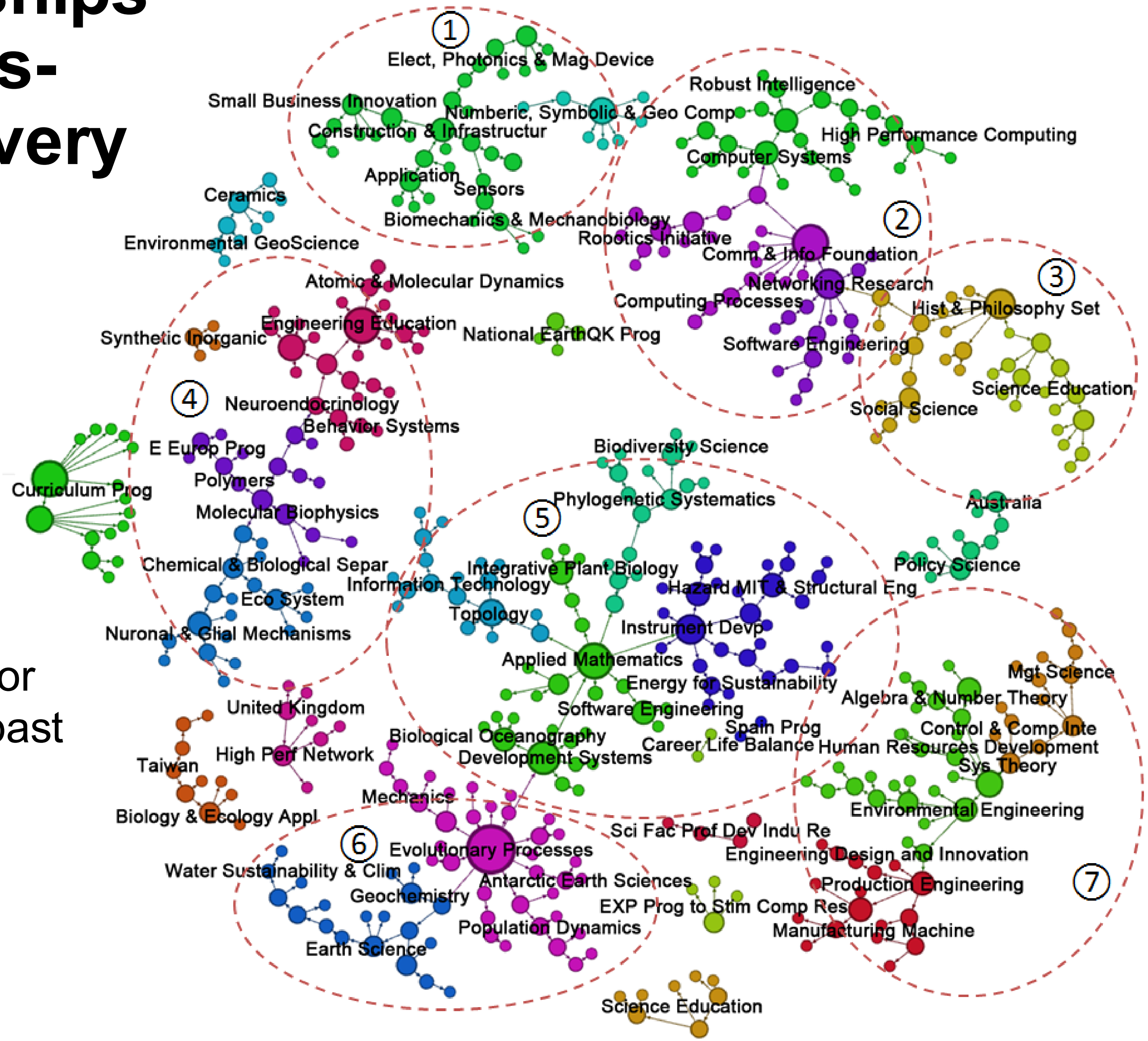
Tracing Topical Relationships - Streaming data analytics-based relationship discovery

A case study of analyzing US NSF proposals

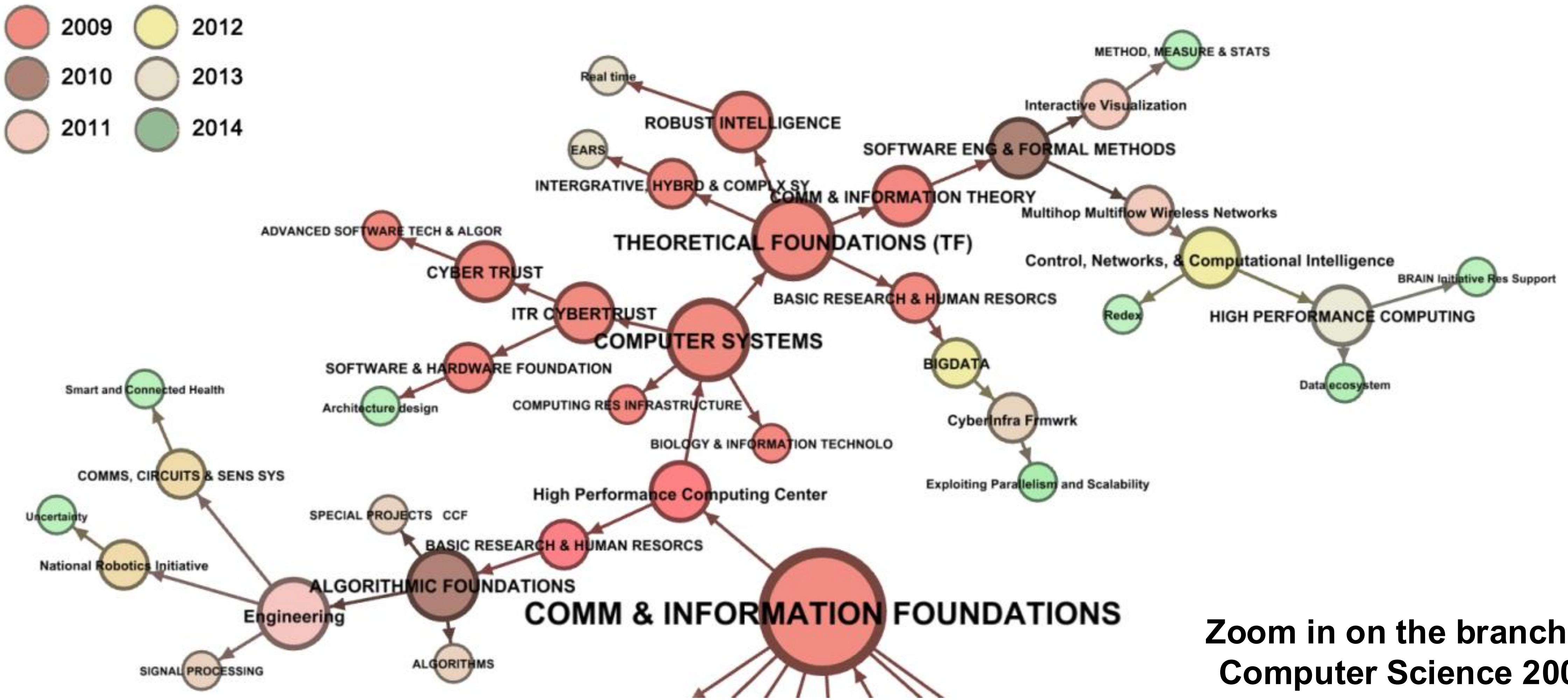
243K academic proposals granted by the United States National Science Foundation between 1976 and 2016

How innovative ideas evolved over time?

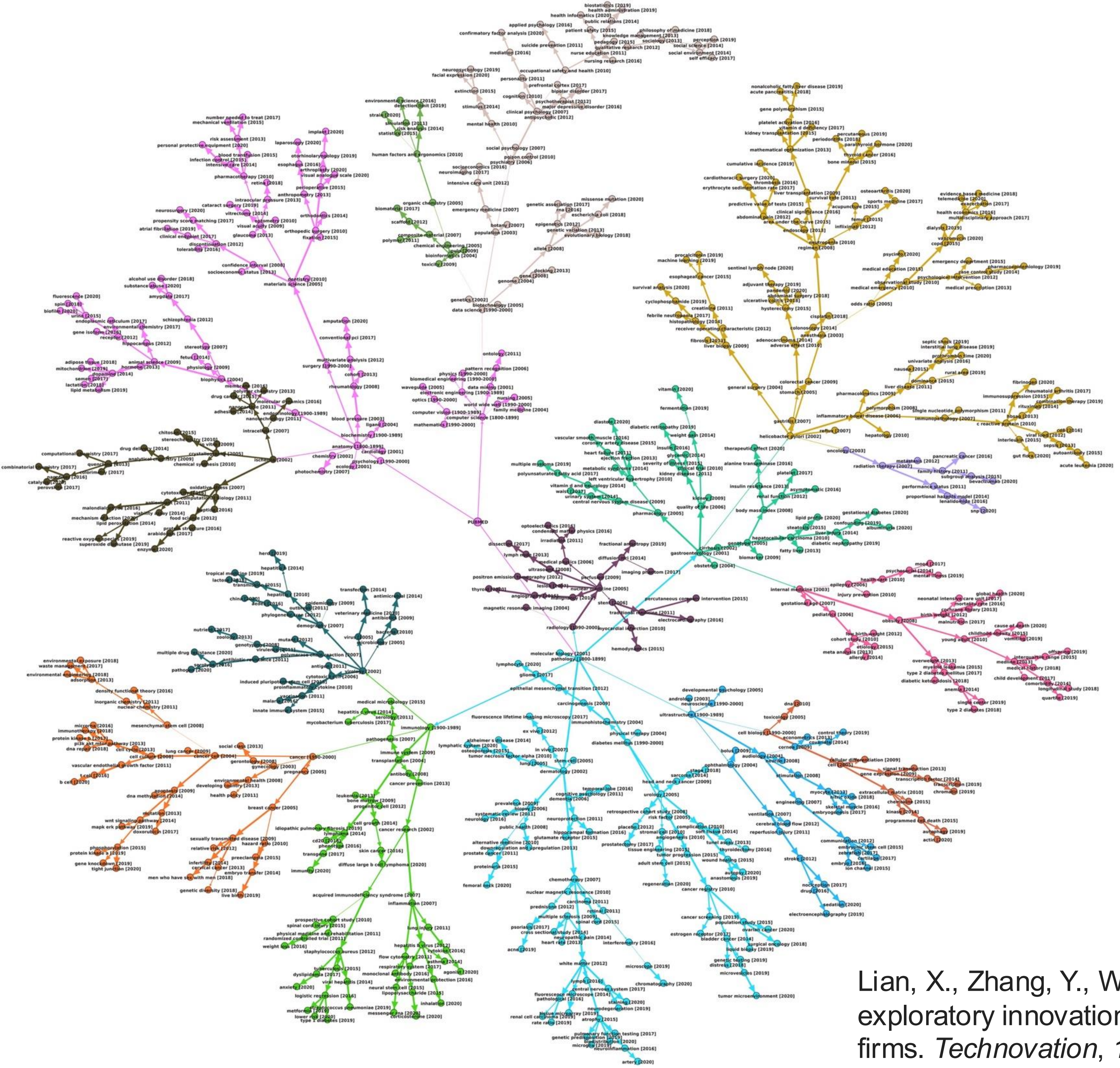
- A research landscape to uncover the major areas funded by the US NSF during the past three decades.
- How inter-/multi-/cross-disciplinary interactions occurred.



Tracing Topical Relationships
- Streaming data analytics-based relationship discovery



Zoom in on the branch of Computer Science 2009-2014



The evolutionary pathways of the entire PubMed dataset

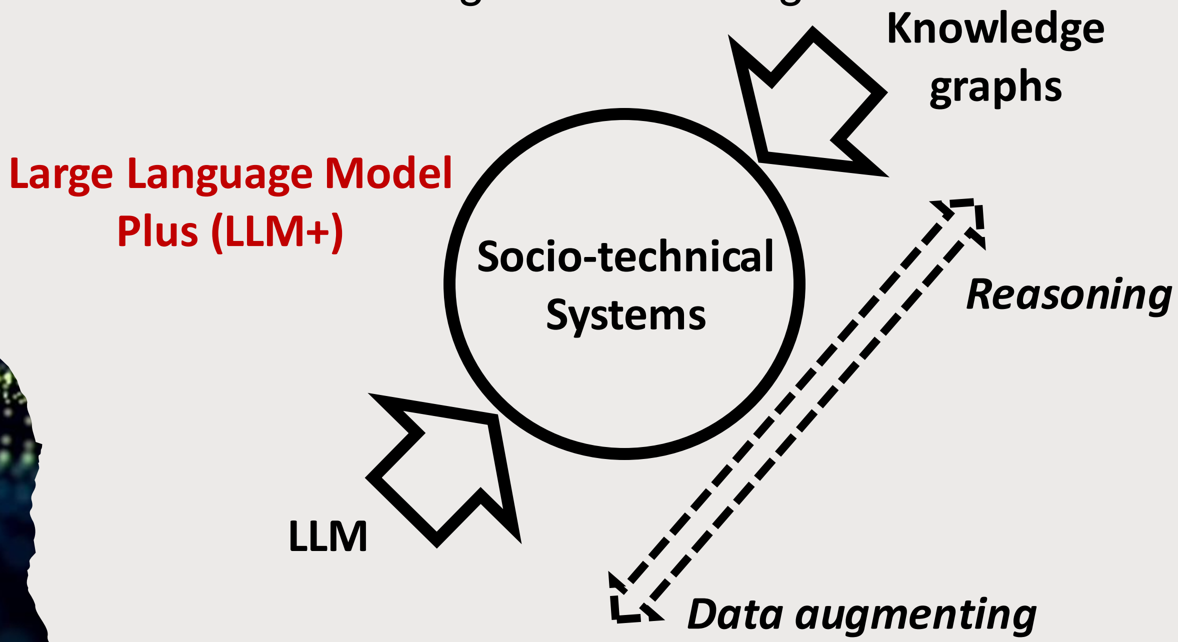
28,330,603 research articles
Timeline: 1800 - 2020
500,259 field of study tags (MAG)

The SEP comprises 716 topics and their predecessor-decendant relationships

Lian, X., Zhang, Y., Wu, M., & Guo, Y. (2025). Do scientific knowledge flows inspire exploratory innovation? Evidence from US biomedical and life sciences firms. *Technovation*, *140*, 103153.

How can LLM support scientometric studies

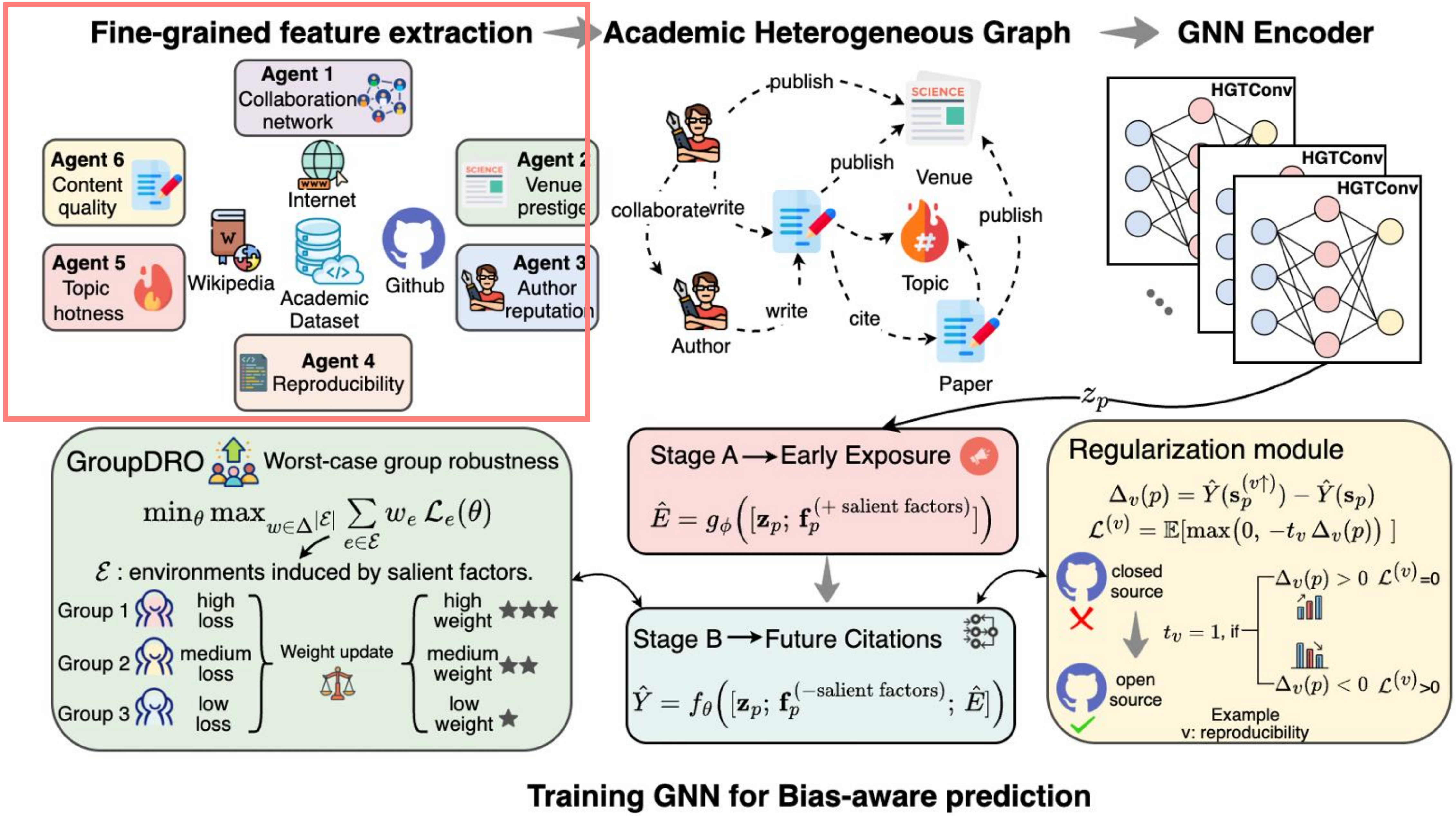
- **Data processing & augmentation**
  - Text summarisation
  - Generating extra features
  - Labelling topics
- **Representation & reasoning**
  - Feature embedding
  - Relationship extraction
- **Prediction**
  - Technological forecasting

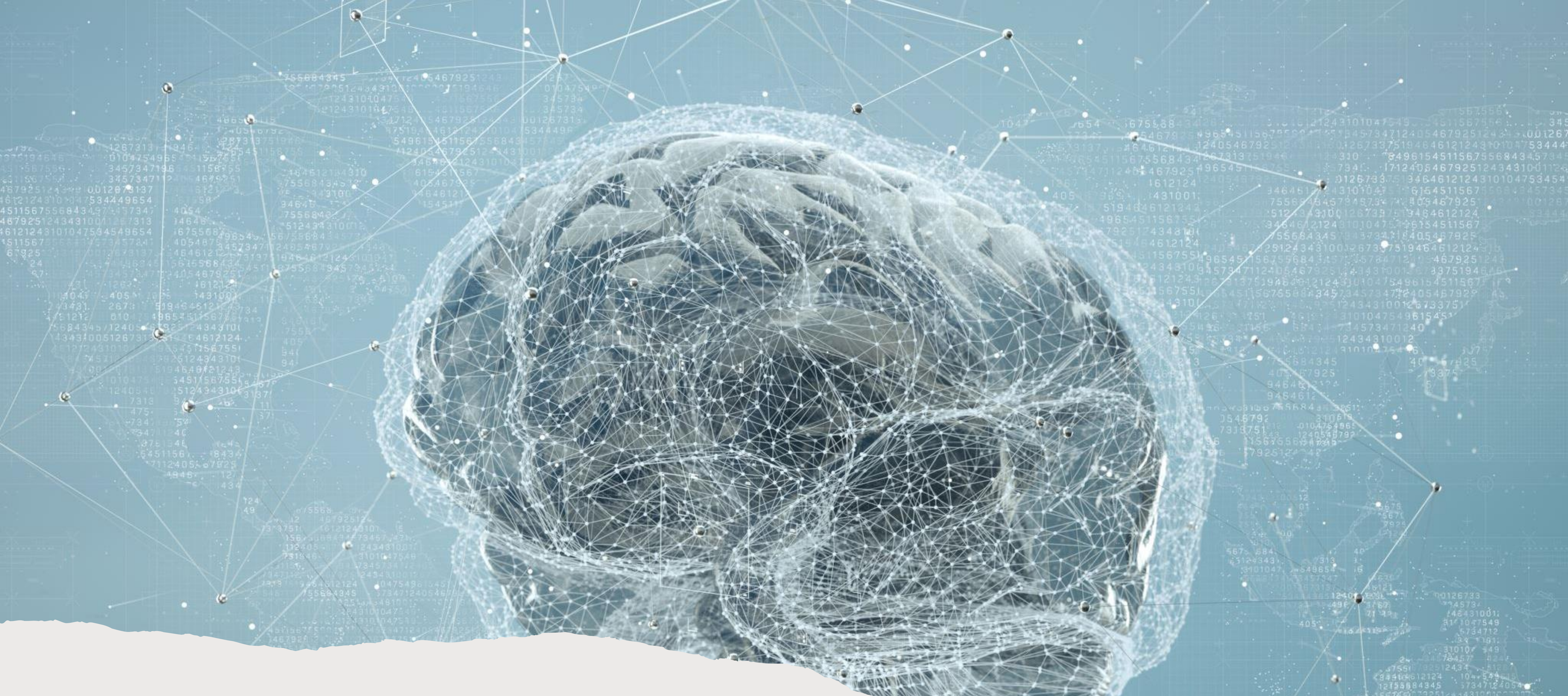


Data Augmentation

Lu, M., Wu, M., Xu, J., Li, W., Liu, F., Ding, Y., ... & Zhang, Y. (2025). From Newborn to Impact: Bias-Aware Citation Prediction. *arXiv preprint arXiv:2510.19246*.

LLMs for feature extraction





Classification

Wu, M., Sivertsen, G., Zhang, L., Qi, F., & Zhang, Y. (2025). Scaling research aim identification: Language models for classifying scientific and societal-oriented studies. *Journal of the Association for Information Science and Technology*, 76(11), 1470-1487.

Language Model-based Classification for Research Aims

Wu, M., Sivertsen, G., Zhang, L., Qi, F., & Zhang, Y. (2025). Scaling research aim identification: Language models for classifying scientific and societal-oriented studies. *Journal of the Association for Information Science and Technology*, 76(11), 1470-1487.

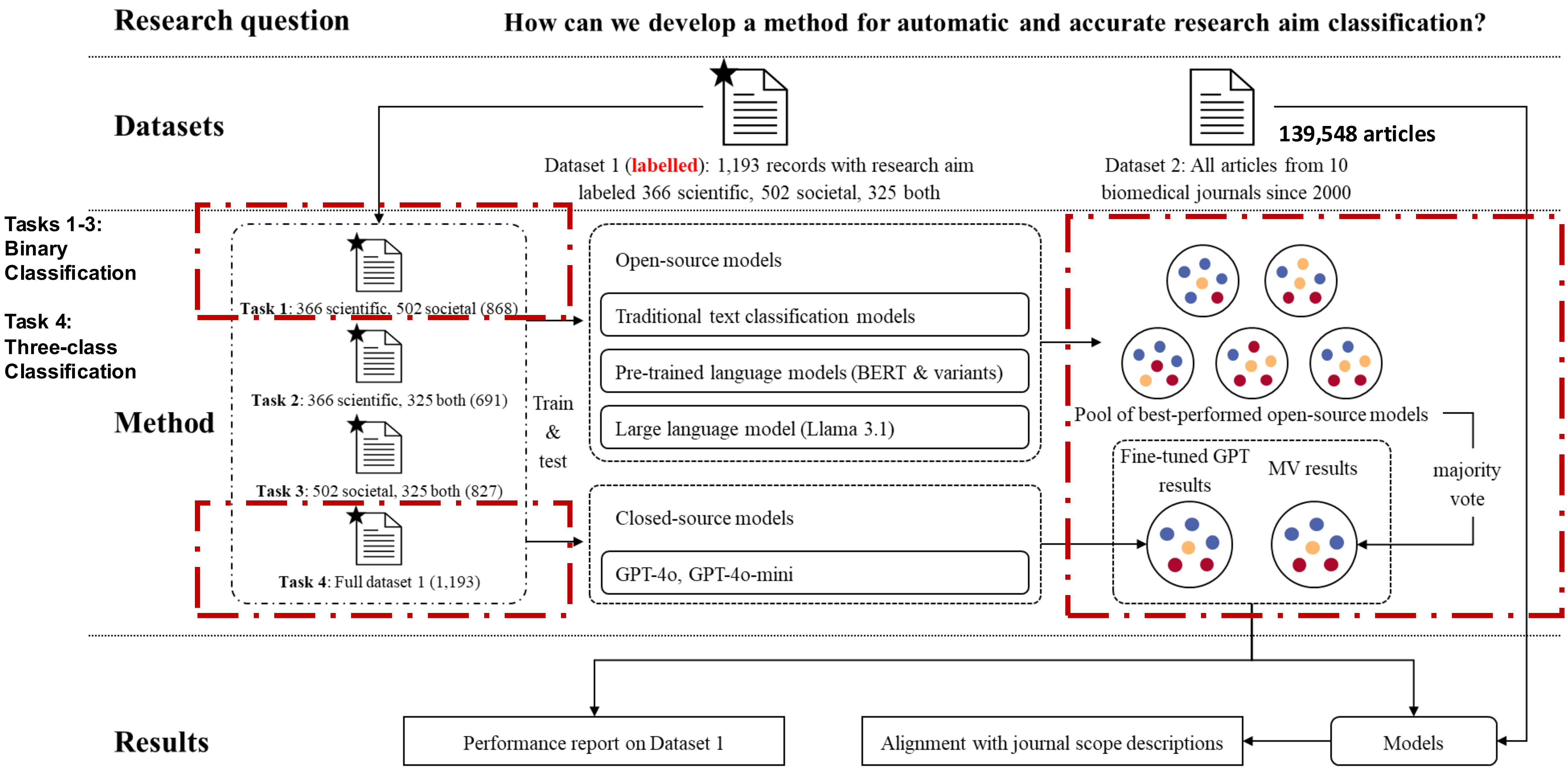
Motivation

What is the benefit of the knowledge or what motivates the knowledge discovery?

- **Scientific Progress: To achieve knowledge advancements by filling gaps and crossing boundaries.**
- **Societal Progress: To achieve external usefulness.**

- Zhang, L., Sivertsen, G., Du, H., Huang, Y., & Glanzel, W. (2021). Gender differences in the aims and impacts of research. *Scientometrics*, 126, 8861-8886.

Research Framework



Binary Classf. on *scient* vs. *societ*

Method	A	Scientific Progress			Societal Progress		
		P	R	F1	P	R	F1
TF-IDF-MNB	0.715	0.977 (1)	0.336	0.498	0.672	0.994 (1)	0.801
TF-IDF-LR	0.805	0.912 (3)	0.599	0.721	0.766	0.959 (2)	0.85
TF-IDF-SVC	0.835	0.849	0.743	0.791	0.827	0.905	0.864
WF-MNB	0.817	0.821	0.725	0.769	0.814	0.883	0.847
WF-LR	0.826	0.813	0.765	0.787	0.834	0.873	0.853
WF-SVC	0.81	0.785	0.759	0.77	0.827	0.849	0.837
Annif-SnB-SVC	0.851	0.857	0.781	0.816	0.849	0.905	0.875
Annif-SnB-FastText	0.79	0.749	0.759	0.753	0.821	0.814	0.817
Annif-SnB-Omikuji	0.838	0.843	0.758	0.796	0.836	0.897	0.864
Annif-S-SVC	0.84	0.856	0.748	0.797	0.831	0.909	0.867
Annif-S-FastText	0.748	0.707	0.69	0.697	0.776	0.791	0.783
Annif-S-Omikuji	0.836	0.867	0.727	0.788	0.821	0.919	0.866
Annif-SL-SVC	0.843	0.842	0.776	0.806	0.844	0.894	0.868
Annif-SL-FastText	0.779	0.724	0.77	0.745	0.823	0.786	0.803
Annif-SL-Omikuji	0.832	0.842	0.744	0.787	0.827	0.898	0.86
F. BERT	0.862	0.863	0.8	0.829	0.86	0.909	0.883
F. SciBERT	0.849	0.862	0.77	0.81	0.845	0.91	0.874
F. BioBERT	0.864	0.858	0.807	0.831	0.868	0.905	0.885
F. SentenceBERT	0.861	0.876	0.777	0.823	0.851	0.921	0.884
F. SsciBERT-e2	0.874	0.87	0.826	0.847	0.877	0.911	0.893
F. SsciBERT-e4	0.881 (2)	0.871	0.846	0.856 (2)	0.889 (3)	0.91	0.899 (3)
F. SsciSciBERT-e2	0.88 (3)	0.891	0.814	0.85 (3)	0.874	0.928	0.9 (2)
F. SsciSciBERT-e4	0.878	0.903	0.794	0.844	0.864	0.937	0.899 (3)
F. RoBERTa	0.858	0.861	0.8	0.823	0.864	0.9	0.88
F. RoBERTa-L	0.866	0.888	0.785	0.83	0.857	0.927	0.889
F. Llama3.1-8B	0.837	0.837	0.774	0.791	0.851	0.886	0.864
ZS GPT4o	0.533	0.473	0.962 (1)	0.634	0.889 (3)	0.22	0.352
FS GPT4o	0.707	0.601	0.911 (2)	0.723	0.893 (2)	0.558	0.687
F. GPT4o-mini	0.919 (1)	0.933 (2)	0.867 (3)	0.899 (1)	0.911 (1)	0.957 (3)	0.933 (1)

Method	A	Scientific Progress			Societal Progress			Both		
		P	R	F1	P	R	F1	P	R	F1
TF-IDF-MNB	0.526	**0.864 (1)**	0.329	0.472	0.48	**0.987 (1)**	0.645	**0.553 (3)**	0.041	0.075
TF-IDF-LR	0.629	0.699	0.664	0.675	0.616	0.834	0.707	0.546	0.283	0.369
TF-IDF-SVC	0.629	0.681	0.686	0.681	0.663	0.758	0.706	0.479	0.371	0.414
WF-MNB	0.637	**0.742 (2)**	0.648	0.688	0.681	0.703	0.692	0.482	**0.528 (1)**	**0.5 (1)**
WF-LR	0.607	0.649	0.645	0.645	0.659	0.713	0.685	0.454	0.403	0.427
WF-SVC	0.595	0.636	0.641	0.637	0.652	0.679	0.665	0.446	0.417	0.43
Annif-SnB-SVC	0.617	0.685	0.659	0.664	0.646	0.754	0.694	0.473	0.374	0.414
Annif-SnB-FastText	0.553	0.523	0.787	0.615	0.631	0.705	0.66	0.271	0.076	0.11
Annif-SnB-Omikuji	0.614	0.685	0.655	0.661	0.646	0.754	0.694	0.469	0.37	0.41
Annif-S-SVC	0.606	0.685	0.642	0.655	0.636	0.749	0.686	0.452	0.358	0.395
Annif-S-FastText	0.536	0.494	**0.788 (3)**	0.591	0.628	0.695	0.65	0.119	0.035	0.054
Annif-S-Omikuji	0.599	0.668	0.649	0.65	0.631	0.732	0.676	0.451	0.353	0.392
Annif-SL-SVC	0.608	0.684	0.651	0.66	0.639	0.738	0.683	0.452	0.371	0.405
Annif-SL-FastText	0.551	0.525	0.782	0.613	0.628	0.701	0.656	0.36	0.079	0.112
Annif-SL-Omikuji	0.61	0.675	0.658	0.659	0.645	0.74	0.687	0.462	0.371	0.408
F. BERT	0.615	0.716	0.678	0.695	0.652	0.786	0.709	0.396	0.288	0.329
F. SciBERT	0.628	0.694	0.691	0.688	0.685	0.748	0.714	0.441	0.383	0.407
F. BioBERT	0.64	0.719	0.701	**0.709 (2)**	0.693	0.737	0.713	0.465	0.425	0.439
F. SentenceBERT	**0.642 (3)**	0.697	0.696	0.695	0.668	**0.843 (3)**	**0.745 (1)**	0.455	0.28	0.343
F. SsciBERT-e2	**0.648 (2)**	0.701	0.688	0.693	**0.713 (3)**	0.747	**0.727 (3)**	0.481	**0.458 (3)**	**0.465 (2)**
F. SsciBERT-e4	0.64	0.704	0.695	**0.699 (3)**	0.71	0.728	0.718	0.458	0.448	0.45
F. SsciSciBERT-e2	0.631	0.699	0.65	0.671	0.695	0.733	0.713	0.46	**0.46 (2)**	0.457
F. SsciSciBERT-e4	0.641	**0.725 (3)**	0.646	0.68	0.693	0.762	0.725	0.476	**0.458 (3)**	**0.461 (3)**
F. RoBERTa	0.629	0.722	0.653	0.685	0.673	0.785	0.723	0.426	0.367	0.391
F. RoBERTa-L	0.573	0.574	0.478	0.52	0.591	**0.884 (2)**	0.695	0.336	0.184	0.201
ZS GPT4o	0.402	0.349	**0.964 (1)**	0.51	**0.745 (1)**	0.229	0.35	**0.561 (2)**	0.04	0.072
FS GPT4o	0.525	0.441	**0.904 (2)**	0.592	**0.719 (2)**	0.508	0.594	0.439	0.115	0.178
F. GPT4o-mini	**0.668 (1)**	0.712	0.744	**0.722 (1)**	0.693	0.798	**0.74 (2)**	**0.567 (1)**	0.389	0.446

Task 4
Triple-class classification:*sci*-
***entific* vs.**
societal* vs. *both

Majority Voting Strategies

- **A trade-off between performance enhancement and cost**, given fine-tuned GPT-4o-mini has the best overall accuracy (but not as impressed as it in the binary classification task
- To integrate the complementary strengths of individual **open-source models** in different classification tasks (Tasks 1-4)
- Two Strategies
  - MV1: Selecting the **five best-performing open-source models** across Tasks 1-4
  - MV2: Systematically evaluating all **possible model combinations** and identifying an optimised configuration

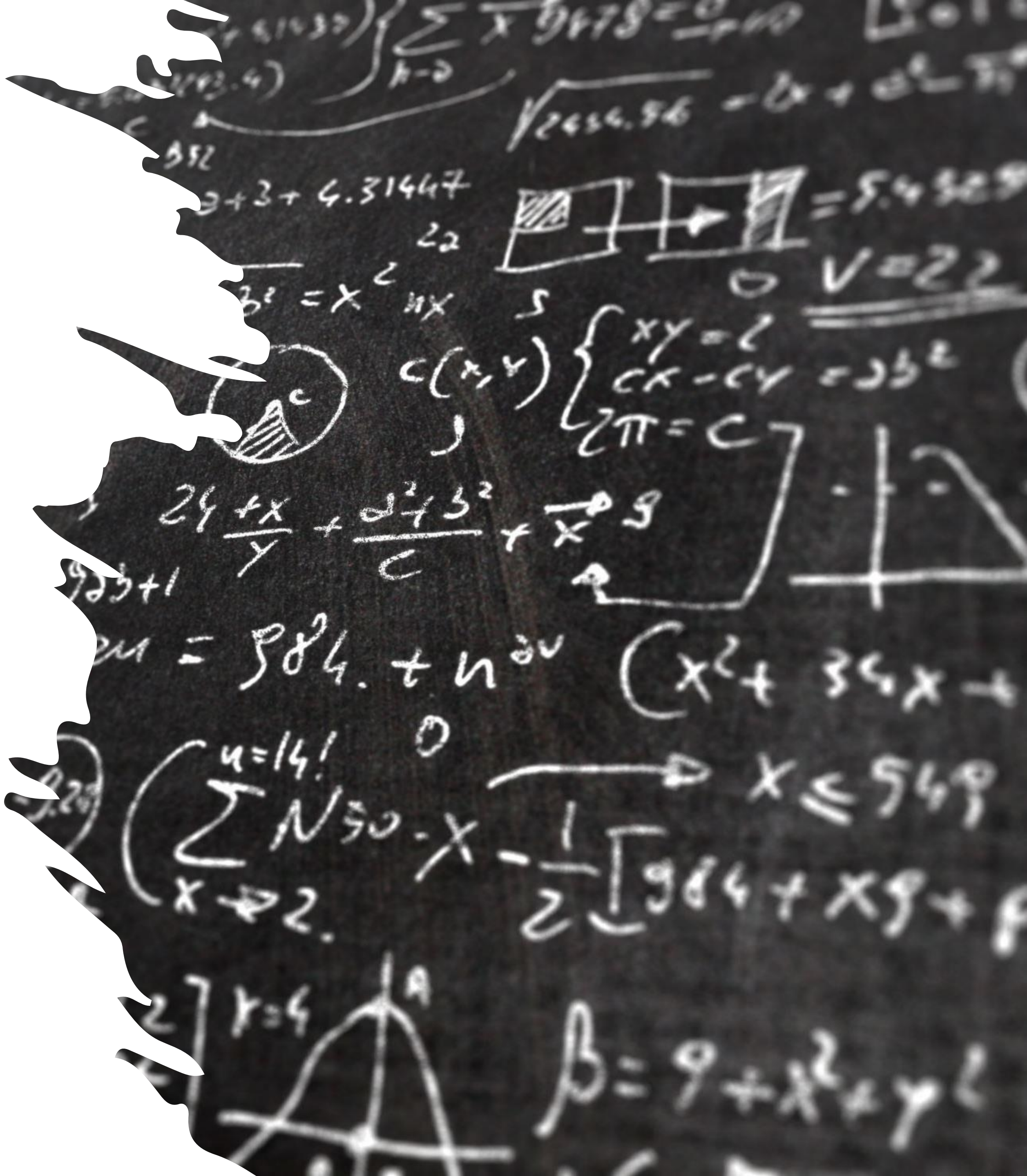
Three-class classification results on Dataset 1 using majority voting strategies

Method	A	Scientific Progress			Societal Progress			Both		
		P	R	F1	P	R	F1	P	R	F1
MajorV1	0.670	0.747	0.683	0.712	0.701	0.813	0.751	0.52	0.443	0.473
MajorV2	0.676	0.750	0.712	0.729	0.702	0.806	0.750	0.524	0.437	0.474

- Both strategies achieved higher accuracies than the fine-tuned GPT-4o-mini model
  - Aggregating results from multiple models can mitigate limitations inherent in individual ones
- A cost-effective alternative without compromising accuracy
- The highest accuracy (below 0.70) somehow coincides with the inter-annotator agreement among human experts (0.74 in Zhang et al., 2021).
  - This indicates that the overall classification performance has reached a relatively high level within the constraints of human-level consistency.
  - Human experts do not fully agree with each other on this manual classification

Future Directions
- Human-AI Collaboration

- **Relationships**
  - Collaborators vs. Tools
- **Reliability**
  - Do you really trust AI?
- **Complementarity**
  - Cross-validation
- **The Sword of Damocles**
  - AI researchers vs. LLM end-users



Thank You!

Yi Zhang (Yi.Zhang@uts.edu.au)